

MUHAMMAD ARFAT

Backend & Secure Systems · Islamabad, Pakistan

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Education

FAST NUCES Islamabad

2021 – 2025

Bachelor of Science in Computer Science – Dean's Honor List

Islamabad, Pakistan

Relevant Coursework

- Parallel and Distributed Computing
- Data Structures
- Database Systems
- Operating Systems
- Computer Networks
- Software Engineering
- DevOps
- Information Security

Experience

SageTeck

Jul 2024 – Present

Full Stack Software Developer

Islamabad, Pakistan

- Engineering backend systems for a multi-tenant hospitality SaaS platform on **Microsoft Azure** – authentication, service orchestration, deployment pipelines, and tenant-isolated data flows across distributed services.
- Customized **SuiteCRM v7 and v8** modules across four enterprise CRM products using PHP, REST APIs, and custom logic hooks; built role-based access controls, scheduler jobs, bean manipulation, and module extensions supporting core business logic.
- Built backend infrastructure for a cloud-based digital signage platform serving **1000+ screens** across WebOS, Android, and Windows; developed the CMS, the WebOS player application, and the per-screen licensing and subscription system.
- Designed and optimized MySQL schemas for high-volume CRM and SaaS deployments, wrote advanced SQL queries, and managed data migrations across staging and production environments.
- Owned GitHub-based deployment workflows, code versioning, and release management across XAMPP, staging, and production environments.

Dropella.io

2021 – 2022

API Development Engineer

Islamabad, Pakistan

- Built the Product Database and Sales Tracking APIs at the core of an e-commerce analytics platform using **FastAPI**.
- Implemented multi-threaded request handling that cut API response times by **~30%** at peak load, improving platform scalability.
- Integrated APIs with front-end systems and third-party services for reliable real-time data flow.
- Conducted code reviews and mentored junior developers on API design and Python best practices.

FAST NUCES Islamabad

2021 – 2022, 2025

Teaching Assistant – Database Systems & DevOps

Islamabad, Pakistan

- Ran labs and graded assignments for undergraduate Database Systems and DevOps & Cloud Native courses.
- Guided students on database design fundamentals, Docker, Grafana, CI/CD pipelines, and cloud infrastructure tooling.

Systems Limited

Jul 2024 – Aug 2024

Intern – Data Analysis

Islamabad, Pakistan

- Worked on database concepts, data warehousing architecture, and Power BI dashboards for business reporting.

Rex Media Group

2021 – 2022

WordPress Developer

Islamabad, Pakistan

- Built and maintained WordPress sites with custom themes, responsive layouts, and client-specific integrations.

Projects

IntraSec – Network Vulnerability Assessment Tool | *Python, Golang, ElasticSearch, Active Directory, Microservices* **2024**

- Built a client-less network vulnerability assessment tool with real-time asset inventory and risk analysis, deployed using a microservices architecture for scalability and modularity.
- Integrated **Active Directory** for asset and identity management; used **ElasticSearch** for indexing and real-time risk querying across the asset inventory.
- Designed the architecture and core scanning engine in Python and Go, with each microservice independently deployable and observable.
- Supervisor: **Dr. Muhammad Asim, HOD Cyber Security, FAST-NU Islamabad.**

CTFd First Blood Challenges Plugin | *Python, Flask, JavaScript, SQLAlchemy* **2021**

- Developed a custom CTFd plugin under contract for **Trillium Infosec**, deployed at the **Ignite Pakistan** cybersecurity competition.
- Implemented a parabolic decay scoring function adjusting challenge points dynamically based on solve count.
- Integrated with CTFd's challenge engine to support hybrid scoring models, improving competitive fairness in CTF events.

LostNFound System | *Java, MongoDB, REST APIs* — GitHub **2021**

- Built a backend solution to manage lost-and-found items in institutional settings, with real-time notifications and user data management.
- Designed for scalable operation across multiple departments with API-driven integration.

YTS.io Scraper | *Python* — GitHub **2020**

- Built an automated scraper for movie metadata and subtitles using Python with API integration and robust error handling.

Technical Skills

Languages: Python, PHP, JavaScript, Go (Golang), Java, C/C++, SQL, HTML/CSS

Backend Frameworks: FastAPI, Flask, Django, CodeIgniter, SuiteCRM

Databases: MySQL, MongoDB, ElasticSearch, Redis

Cloud & DevOps: Microsoft Azure, Docker, Nginx, CI/CD, Apache Kafka, Git, GitHub Actions

Security & Systems: REST API Design, Microservices, Active Directory, Vulnerability Assessment, Multi-tenant SaaS, Role-Based Access Control

Honors & Awards

Dean's Honor List – FAST NUCES Islamabad, Fall 2024.

Streamlit Student Ambassador – FAST NUCES Chapter, May 2023 – May 2024.